

Inside Out 2 characters: revisiting the old and making space for the new

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Figure 1: The full cast of emotion characters in *Inside Out 2* ©Pixar

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Abstract

Inside Out 2 (2024) presented a unique set of technical and artistic challenges, requiring careful decisions on retaining elements from the original film while embracing new technologies. As we transitioned from RenderMan's REYES to the newer RIS framework, we navigated the complicated task of preserving the iconic look, particularly the emotions and human elements. We enhanced the visual quality and accessibility for artists through advancements in our

character pipeline, including reusing shading libraries and optimizing procedural workflows in Houdini, implementing show emotion specific hair illumination. Innovative rigging and grooming techniques were created to address technical and design challenges in character creation. Lighting techniques were refined to reflect the characters' story arcs, we also enhanced the Hexport [Coleman et al. 2020] system to be more user-friendly and accessible, streamlining shading and lighting workflows while reducing computation time and storage costs.

CCS Concepts

• **Social and professional topics;** • **Computing methodologies**
→ **Computer graphics;**

Keywords

Inside Out 2, Pixar, Characters, Shading, Lighting, Rigging

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1 Introduction

When we began production on *Inside Out 2*, we faced a unique set of technical and artistic challenges. Nearly a decade had passed since the release of the original film, during which our pipeline has significantly evolved, and advancements in rendering technology have emerged. With access to the original files and artists that worked on *Inside Out*, we were tasked with carefully deciding what elements to retain from the first film and which new techniques to embrace, ensuring that the sequel would expand on the worlds, while taking full advantage of the latest innovations.

2 Challenges and Implementations

Art Direction and Complexity. Since *Inside Out 2* is a sequel, we had the advantage of knowing what the look of the original characters were and how they were constructed [Angelidis et al. 2015]. We also knew what worked well and what needed to change for this film, both technically and design wise. Early on in the film, we established with our Art Department the enhancements we wanted to implement for the core 5 emotions. These improvements informed our choices for the new cast of characters.

RIS VS REYES. The original film was developed using RenderMan's REYES [Cook et al. 1987] renderer. In REYES, RSL shaders were very flexible, with full control of the illumination model and arbitrary messaging between surfaces and lights. This made non-physical behaviors easy to implement and control. Shader evaluations were cached and reused to minimize costs. Lighting rigs were heavily used and creating the scenes was an involved process. RIS, *Inside Out 2* ray-trace based renderer, is physically based and designed for effective global-illumination. Scenes composed of physically based shaders are simple to set up and control. Since shaders in RIS are not cached as in REYES, they need to be optimized for better performance. We encountered a number of challenges

re-implementing some of the non-physically based behaviors in RIS which we tackled by leveraging the strengths of the new pipeline and renderer.

Emotions and Humans. The original film utilized procedural primitives for features like hair dots and skin particles, which we re-implemented in Houdini with the help of our Hexport [Coleman et al. 2020] system. This shift improved inspection, debugging, and introduced more dynamic behaviors. We also refined the emotions' design elements, such as skin particles and volumetric glows, to enhance their appearance in the new framework. For Riley's character, we adapted skin shading from "Turning Red," adding subtle blemishes to emphasize her humanity, and differentiate her even more from the mind world. Our updated shading pipeline, which allows for re-use of shading libraries, streamlined the process while enabling us to preserve the production's unique style.

The emotions' skin is covered in a layer of randomly displaced, pulsing particles, and has a light response that is broadly surface-like, with higher frequency-per-particle variation to give a textural quality. We used our Hexport [Coleman et al. 2020] system to generate the skin particles in a Houdini pre-pass, making it easier to implement more complex particle behavior, including rigging-driven deformations and dynamic de-intersection with hair and garments. The Emotions are surrounded by layers of emissive auras that convey an ethereal "life force". The aura algorithms were designed to be efficient and temporally coherent for global illumination. We derived the auras from volume field arithmetics, combining weighted solid angle fields and density fields induced from camera ray transmission, combined with a voxel resolution heuristic derived from human perception of the human body. To shape the auras, we developed a system that is reusable and editable across characters in minutes. Joy's design called for her to be soft and translucent near her silhouette edges, with a more solid and opaque core. During the production of *Inside Out* this was achieved by rendering her body using a heterogeneous volume with a density falloff near the surface of the underlying model, and a traditional surface shading model extrapolated to the volume. We found it difficult to implement this technique in RIS and instead rendered her body as a traditional subdivision surface with an opacity fall off near the silhouette edges.

Lighting. In the first film, Joy was only lit by her own lights. In *Inside Out 2* we pushed the lighting by giving Joy the ability to react to the lights in the scene as well. For *Inside Out 2* we wanted to maintain her familiar look, while also visualizing Joy's story arc by seeing her react to her environment and the objects she interacts with. Since she is emissive and bright, we needed to darken her glow lobe to the lighting response without going far overrange. We used AOVs of individual lights as masks in Nuke to dim and tint her emissive signal and provide space for other lights to add to her illumination. Our inherited Katana rigs aided this process.

Groom Advancements. The emotion's hair is made out of dots that are aligned along the strands. They have a sparkly and ethereal quality while still retaining the hair-like behavior. For the emotions in the original film, we used a render procedural that created ray facing disks along the hair strands to implement the hair dots. The disks faced the shadowcameras creating natural looking shadows.



Figure 2: Riley, Bree and Grace in *Inside Out 2* ©Pixar

For our show we no longer have discs that can align with shadowmaps, so we created a houdini engine set of discs and spheres that provided the same functionality in the new renderer. We combined two instances of the Chiang hair model [Chiang et al. 2015], with added diffuse and emissive lobes. We integrated the MaterialX Lama system, enabling artists to break energy conservation rules to achieve non-photorealistic looks. We need the emotions of other humans besides Riley to be easily recognizable by the audiences. We achieve this by giving the emotions corresponding hairstyles with their human counterpart. Rather than recreate each complex hairstyle up to nine times each, we developed a process that allowed us to refit the human hair onto the various emotion heads. This process used curvenets [de Goes et al. 2022], a light, topologically independent prim that provides a level of automation while retaining the ability to fine tune and direct the shapes.

Rigging Innovation. We also developed new features to support the specific needs of our new characters and provide animators with a structured workflow that includes controls from rough to fine. Anxiety, one of our new emotions, has a distinctive character design with a horizontal head shape and large, expressive eyes. Specialized face rigging was essential, especially in the mouth area where supporting bones extend from the neck, which contrast with flexible cheek pockets. We implemented a layered control system allowing both broad control and precise mouth manipulation: one layer for the outer cheek shape, one layer of detailed adjustments for the mouth corners to move on the surface, and one layer for fine-tuning polish using curvenet [de Goes et al. 2022]. Additionally, controls for wavy lip shapes and a synchronized teeth rig were incorporated. These rigging solutions provide animators with a comprehensive toolkit to create Anxiety’s wide range of expressions and movements.

Computing and Sharing Character Data. Asset production for *Inside Out 2* occurred alongside the streaming series for Disney+,

with both projects sharing many characters and assets. To streamline this process, we developed a new system for live asset sharing across multiple parallel productions. This system allowed artists to make creative changes in one production’s working area, instantly making those updates available to the other production. This automated sharing reduced the typical overhead of manually copying data between projects. Key shared assets included character models, rigs, shading data, test shots, and crowd setups, all managed efficiently within production environment. Additionally, we enhanced the Hexport [Coleman et al. 2020] system by making it more user-friendly and accessible, which streamlined shading and lighting workflows and reduced computation time and storage costs.

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